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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2016-2017

CE2005/CZ2005 – OPERATING SYSTEMS

Apr/May 2017 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 7 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. Questions **DO NOT** carry equal marks.

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1. (a) Indicate whether each of the following statements is true or false. Justify each answer.

(i) The operating system maintains a list of all running processes in a separate running queue.

(2 marks)

(ii) Some I/O instructions do not require a switch to kernel mode.

(2 marks)

(iii) In multi-programming systems, if a job is waiting for I/O, then the CPU will wait for this job.

(2 marks)

(iv) Direct Memory Access eliminates the need for interrupts during data transfer.

(2 marks)

(v) It is possible for a process to transit from the running state to the ready state.

(2 marks)

Note: Question No. 1 continues on Page 2

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(b) Table Q1 lists a set of processes with their respective CPU-burst time and arrival time. P2 and P5 arrive at time 2 in the order of P2 then P5. Calculate the average waiting time when the system uses each of the following scheduling algorithms: Non-preemptive Shortest Job First and Shortest Remaining Time First.

Table Q1

Process	Burst Time	Arrival Time
P1	5	0
P2	4	2
P3	4	4
P4	3	5
P5	2	2

(4 marks)

(c) A ready queue is partitioned into two sub-queues: the foreground queue for interactive processes, and the background queue for batch processes. The foreground queue uses Round-Robin scheduling, while the background queue uses First-Come First-Serve scheduling.

(i) Justify the given choice of scheduling algorithm for each sub-queue.

(2 marks)

(ii) Suggest and explain two different methods for allocating CPU time between the two sub-queues.

(4 marks)

(d) The ready queue is implemented as a singly linked list of Process Control Blocks (PCBs). Two pointers, namely *head* and *tail*, point to the first and last PCB in the list, respectively. When a timer interrupt occurs, the list has three PCBs in the order of PCB1 (first) -> PCB2 -> PCB3 (last). No processes leave or join the queue after the timer interrupt.

(i) Assume that Shortest Remaining Time First scheduling is used. Which is the next process to run on the CPU? Justify your answer.

(2 marks)

(ii) Assume that Round-Robin scheduling is used. Update the linked list's pointers when the next process is chosen to run.

(3 marks)

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2. (a) Indicate whether each of the following statements is true or false. Justify each answer.

(i) Bounded-waiting is violated when a process is in its remainder section, but it still holds a lock into the critical section.

(2 marks)

(ii) There is no problem if the semaphore operation signal() is not atomic.

(2 marks)

(iii) Time-sharing systems do not have deadlock problems.

(2 marks)

(iv) In semaphore implementation, blocking is not always better than busy waiting.

(2 marks)

(v) If the resource allocation graph does not contain cycles, a deadlock must occur in the system.

(2 marks)

(b) Consider the current system's state with three processes (P1, P2 and P3) and three resource types (A, B, and C). The current Allocation and Need matrices are shown in Table Q2b.

Table Q2b

		Available
		A B C
		2 0 1
Process	Allocation	Need
	A B C	A B C
P1	1 3 2	3 1 0
P2	3 1 2	1 0 0
P3	2 0 1	2 1 3

(i) Find all the safe sequences for this system.

(2 marks)

(ii) If P1 requests for (A=2, B=0, C=0), should the request be granted and why?

(2 marks)

Note: Question No. 2 continues on Page 4

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(c) Consider a readers-writers problem, in which there are multiple concurrent readers and writers. A writer requires exclusive access to the database, while multiple readers can read at the same time. To ensure good performance, no more than **four** concurrent readers are allowed in the database. The pseudo-code of an initial solution using semaphores is given in Table Q2c.

Table Q2c

Shared data	Writer process	Reader process
semaphore *rd = 4*, *wr = 1;*	*wait*(*wr*); *do writing;* *signal*(*wr*);	*wait*(*rd*); *wait*(*wr*); *do reading;* *signal*(*wr*); *signal*(*rd*);

(i) Assume blocking implementation is used for the semaphores. Now there are ten blocked processes, including three readers and seven writers, waiting to execute "*do reading;*" and "*do writing;*", respectively. List all possible values for *rd* and *wr* at this time instant. (Hint: consider different cases in which a reader is reading, or a writer is writing, etc.)

(4 marks)

(ii) Identify and explain one issue with the solution given in Table Q2c.

(2 marks)

(iii) Modify the pseudo-code in Table Q2c to ensure that the solution is correct and deadlock-free. You are only allowed to use semaphore operations, conditional statements and addition/subtraction instructions. You may introduce new semaphores or variables if needed.

(5 marks)

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3. (a) Indicate whether the following statements about memory management are true or false. Justify your answers

(i) Physical address space of a process is always contiguous.

(2 marks)

(ii) Memory allocation using fixed partitioning approach limits the degree of multi-programming.

(2 marks)

(iii) Worst-fit dynamic storage allocation policy will produce the smallest leftover hole.

(2 marks)

(iv) Using inverted page table will increase address translation time.

(2 marks)

(v) With virtual memory support, the degree of multi-programming can be increased.

(2 marks)

(b) Consider a demand paging system that uses a fixed allocation policy for frame allocation and Least-Recently-Used (LRU) algorithm for page replacement. Assume logical address has 12 bits, page size is 512 bytes, and there are 16 frames in the physical memory. Figure Q3 shows the page table of a process. A dash in an entry indicates the corresponding page is not in the memory.

Page Number	Frame Number	Time of Use
0	0	1
1	5	3
2	-	-
3	-	-
4	12	4
5	-	-
6	3	2
7	-	-

Figure Q3

(i) What are the logical and physical address formats?

(3 marks)

Note: Question No. 3 continues on Page 6

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(ii) Translate the following logical addresses to the corresponding physical address:

- 010010110101
- 001101001001

(3 marks)

(iii) Assume each page table entry has a time-of-use field. A counter is incremented whenever a page is referenced. Then, the value of the counter is copied to the time-of-use field in the corresponding page table entry. Given the page table shown in Figure Q3 and the current counter value (that is, 4), how many page faults will be generated for the following page reference string?

0, 4, 3, 1, 6, 4, 0, 2

Show entries of the page table after executing the above page reference string.

(8 marks)

(c) What is the cause of thrashing? How does the system detect thrashing? Once it detects thrashing, what can the system do to resolve this problem?

(6 marks)

4. (a) Indicate whether the following statements about file system and I/O are true or false. Justify your answers.

(i) Open file table is used to temporarily cache data blocks of a file to improve efficiency.

(2 marks)

(ii) Both directories and files reside on hard-disk.

(2 marks)

(iii) Using tree-structured directories, a file can have different names.

(2 marks)

(iv) Buffer is used to hold a copy of data item that resides elsewhere.

(2 marks)

Note: Question No. 4 continues on Page 7

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(v) All disk scheduling policies covered in the lectures, except for First-Come-First-Served (FCFS), may have starvation problem.

(2 marks)

(b) Consider a file system on a disk that has both logical and physical block size of 1024 bytes. Assume that the file directory is already in the memory. For each of the following disk space allocation strategies, to access the 5000th byte of a 12 Kbytes file, how many physical blocks must be read from the disk?

- Contiguous allocation
- Linked allocation
- Indexed allocation

(6 marks)

(c) In UNIX 4.0BSD, the size of blocks used in the file system is increased from 512 bytes to 1024 bytes. Discuss its effects on disk space utilization and throughput.

(4 marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.